Speak and Be Known: Authenticating Users via Ear Canal Deformation on Earbuds

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***Abstract*—With the increasing popularity of smart wearable devices, such as earbuds and smart watches, presents new challenges for seamless and secure user authentication due to their limited user interfaces. Conventional biometric methods, including voice, fingerprints, and facial recognition often face issues such as usability limitations, interference from noise, or vulnerability to spoofing attacks. This paper introduces a novel authentication system called BaroAuth, which utilizes the stable and unique Speech-aware Pressure Sequences (SPSs) patterns captured by a miniaturized MEMS barometer embedded in earbuds. The design of BaroAuth hinges on two important observations. First, the production of speech relies on the coordinated movements of articulatory organs, including the tongue, jaw, and soft palate. These organs, through the activity of the temporomandibular joint (TMJ), alter the shape of the ear canal, thereby causing subtle pressure changes that encode the speaker's unique physiological characteristics and articulatory dynamics. Second, SPSs show significant intra-individual consistency and considerable inter-individual variability, which barometers can effectively measure. Meanwhile, we develop the BaroAuth prototype and carry out comprehensive experiments based on it. The experimental findings reveal that BaroAuth demonstrates a mean false-acceptance rate (FAR) of 0.41% and a false-rejection rate (FRR) of 1.23%, respectively, even under complex attack scenarios.**

***Index Terms*—Speaking user authentication, earbuds, ear canal deformation, barometer.**

I. Introduction

Recently, smart wearable devices have been widely applied, with the growth of earbuds being particularly rapid. The global earphone market is projected to grow to $145 billion by 2031, with an annual compound growth rate of 17.0% [1]. In addition to providing convenient communication and entertainment functions, earbuds are gradually developing into a key interaction gateway in the IoT ecosystem. They are playing an increasingly important role in identity authentication in application scenarios such as password-free payment, smart home control, and voice assistants.

User authentication methods for earbuds can generally be divided into three categories. First, activity-based methods authenticate users by extracting dynamic features generated from daily human activities. A prominent approach in this category is voiceprint-based authentication, which identifies individuals based on unique vocal characteristics such as tone, pitch, and rhythm during speech. While voiceprint authentication is commonly used in earbuds due to its ease of integration with microphones, its reliability is often compromised by environmental noise, imitation, and adversarial attacks [2], [3], [4]. Other approaches in this domain exploit facial muscle behavior [5] and acoustic features [6], [7], [8], [9] to support user verification. These methods also maintain usability without user input, but they are often affected by significant body movement artifacts due to the characteristics of Inertial Measurement Unit (IMU) sensors. The second category consists of interaction-based methods, which rely on predefined user actions, such as dental occlusion [10] or specific facial gestures like finger sliding [11], [12]. These methods require active user participation, limiting their practicality in everyday scenarios. Furthermore, repetitive actions are often less reliable, prone to operational errors, and more vulnerable to spoofing attacks. The third category is biometric-based methods, which utilize unique physiological signals, such as breathing [13], heartbeat [14], [15], or ear canal geometry [16], [17]. While these methods offer high uniqueness and improved privacy, they typically rely on ultrasonic modulation, resulting in increased energy consumption and potential interference with the earbuds' normal audio functions. Additionally, concerns about the potential risks of prolonged exposure to ultrasonic signals present challenges for their widespread adoption.

In this paper, we propose BaroAuth, an innovative user authentication system that utilizes speech-induced pressure variations within the ear canals as a robust dynamic biometric. The core concept behind BaroAuth is that *Ear Canal Dynamic Motion (ECDM) [18] results from the intricate interactions of articulatory organs such as the lower tongue, jaw, and oral*

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cavity. This leads to strong consistency within individuals and considerable variability across individuals, positioning ECDM as a promising dynamic biometric [19]. BaroAuth incorporates a miniature MEMS pressure sensor within an earbud housing to continuously and passively sense speech-induced pressure fluctuations. SPSs provide a distinct biometric solution that effectively balances privacy, convenience, and reliability. Their inherent properties guarantee security, while the use of natural speech allows for seamless and user-friendly authentication.

BaroAuth functions in two phases. In the offline enrollment phase, the user utters a pre-defined passphrase repeatedly while BaroAuth records the corresponding SPSs. These samples are then used to build a training dataset for an end-to-end deep learning model, which learns the user-specific patterns required for authentication, thereby removing the need for manual feature extraction. In the online authentication phase, the user speaks the passphrase once, and BaroAuth assesses whether he/she is legal or not.

Challenges and Contributions: The design of BaroAuth encounters three challenges, as follows:

1) Mitigating Baseline Drift: Environmental factors, like changes in altitude and temperature, can lead to baseline drift in pressure signals, which undermines authentication accuracy. To address this issue, we utilize Empirical Mode Decomposition (EMD) to dynamically isolate and eliminate low-frequency components, ensuring a stable and drift-free pressure signal.

2) Capturing Time and Frequency Domain Features Complementarily: SPSs demonstrate how dynamic speech behavior influences the static physiological structure. In addition to changes in mouth shape, vocal cord vibrations and oral cavity resonance also produce subtle pressure fluctuations within the ear canal. While both factors affect authentication accuracy, their complementary features are challenging to capture. To address this, we design an end-to-end deep learning model that seamlessly integrates time-domain features (which reflect mouth shape changes over time) with frequency-domain features (which capture pressure variations due to vocal cord vibrations and oral cavity resonance). This enables the model to effectively capture both global trends and local characteristics, leading to enhanced authentication accuracy.

3) Addressing Variations in Speaking Rhythm and Limited Training Data: Variations in speaking rhythm may reduce the ability of our authentication model to distinguish effectively. To overcome this challenge, we generate pseudo SPS samples by applying time-stretching and compression techniques to corresponding speech signals. This process simulates diverse speaking rhythms and enriches the dataset for improved model robustness. By generating multiple pseudo samples from a limited number of original recordings, we significantly enhance the model's robustness and adaptability to real-world variations.

Our contributions are threefold:

- We introduce BaroAuth, an innovative user authentication method that utilizes dynamic biometric SPSs. BaroAuth delivers high accuracy and exhibits strong robustness in challenging mobile environments (e.g., walking, taking an elevator, and climbing stairs), while also offering significant resistance to speech impersonation attacks.
- We integrate miniature MEMS pressure sensors (Bosch BMP390) into a pair of earbuds to create a BaroAuth prototype, ensuring low power consumption and high hardware compatibility, while preserving the normal audio output functionality of the earphones.
- Extensive real-world experiments are conducted with 25 volunteers to assess the performance of BaroAuth, examining factors such as speaking rhythm, motion scenarios, and command length. The results demonstrate its efficacy and robustness. Additionally, the experimental results demonstrate that, even in the presence of speech impersonation attacks, BaroAuth maintains a low false acceptance rate (FAR) and a false rejection rate (FRR) of 1.62% and 1.74%, respectively.

II. Preliminaries

A. System and Threat Model

BaroAuth utilizes barometers integrated in earbuds to capture SPSs, offering an innovative and secure method for user authentication. The barometers passively and continuously detect pressure variations in the ear canals. BaroAuth focuses on a session-based usage model. Each time the earbuds are put on, it launches a short, on-demand authentication session and prompts the user to speak the passphrase. After a successful attempt, the user can use the earbuds and associated services without further checks until the earbuds are removed and worn again. During registration, users are asked to speak a predefined passphrase, which will be used for future authentication. No special actions or additional learning are required from users during either the registration or authentication process. Users are free to move during speech, including activities like walking, taking an elevator, or climbing stairs.

Our work assumes a threat model in which the attacker has full knowledge of how BaroAuth operates and has obtained relay access to the victim's earbuds. We consider two types of potential attack scenarios:

Scenario-A: Zero-knowledge Attack: The attacker knows the voice command used by the victim for authentication and attempts to mimic it by uttering the same command without detection. However, the attacker has no ability to duplicate the full range of articulatory motions that the victim employs to produce the command.

Scenario-B: Knowledge-based Attack: This attack scenario mirrors the zero-knowledge attack process but assumes that the attacker covertly observes or records the victim's manner of speaking the target command. Through prolonged observation and analysis, the attacker gains proficiency in mimicking the victim's articulatory motions.

B. Speech-Aware Pressure Sequence

Speech is generated through the coordinated actions of articulatory organs such as the jawbone, tongue, and lips, with support from the temporomandibular joint (TMJ), which links the mandible to the temporal bone. These movements dynamically modify the vocal tract, thereby significantly affecting the

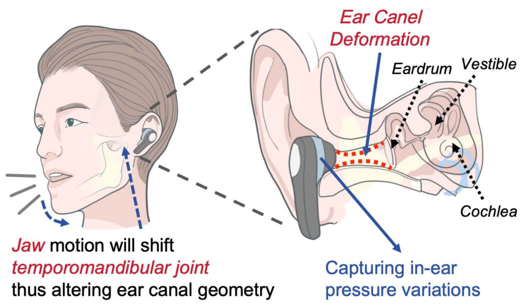


Fig. 1. Speech-induced jaw motion shifts the TMJ, resulting in deformation of the ear canal and changes in inner ear pressure, captured for user authentication.

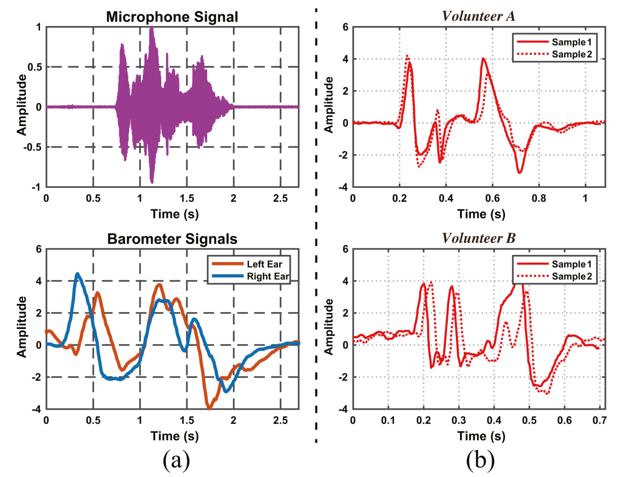


Fig. 2. (a) As a speaker says "Navigate to my office", the microphone captures the raw audio data; meanwhile, the barometer records the corresponding SPS. (b) Pressure signals from left ear sampled during volunteer A (above) and B (below) uttering command "Video my mom" twice.

geometry of the ear canals, a phenomenon referred to as ear canal dynamic deformation. Primarily driven by jaw and tongue activity, these deformations lead to changes in the ear canal's geometric properties, such as volume and diameter (as illustrated in the Fig. 1). For example, jaw movements can result in ear canal volume changes between 10 mm^3 and 25 mm^3 [19], or diameter alterations up to 2.5 mm [20], depending on the intensity and direction of the movement. Although such deformations show considerable inter-individual variability, they exhibit strong consistency within individuals, reflecting unique anatomical and behavioral characteristics. As such, ear canal dynamic deformation presents a promising biometric feature for user authentication.

Definition 1: A **Speech-aware Pressure Sequence** refers to the dynamic deformation of the ear canal caused by speech production. It comprises two main components: 1) The immediate movements of articulatory organs, including the jaw and tongue, which drive TMJ movements and directly result in geometric changes in the ear canal, such as expansion or compression; 2) The secondary deformation of the ear canal geometry induced by traction. These deformations occur due to the coordinated actions of muscles like the masseter, zygomaticus, and sternocleidomastoid, which reshape the ear canal's geometry through traction effects.

C. Distinguishability of SPSs

In contrast to active acoustics-based ear canal deformation detection methods, we have developed Barobuds, an in-ear earbuds equipped with pressure sensors (see Fig. 10(a). Barobuds passively capture real-time dynamic pressure variations in the ear canals, enabling robust speaker authentication. As illustrated in Fig. 2(a), when a participant says "Navigate to my office," pressure signals from both the left and right ear canals precede the audio signals recorded by a microphone. This occurs due to preparatory actions, such as jaw adjustment and mouth opening before vocalization, as well as post-speech movements like jaw retraction and mouth closure, which introduce a delay in the pressure signals. These pressure changes reflect articulatory patterns, revealing unique speaking habits and physiological traits. Even if an attacker mimics the victim's articulatory motion, structural differences in the ear canal lead to distinct pressure patterns, ensuring reliable authentication.

Furthermore, data analysis from multiple participants shows consistent differences in dynamic pressure between the left and right ear canals of the same individual, once variations in device performance are excluded. This discrepancy is likely due to asymmetric facial muscle exertion during speech or anatomical differences in the ear canals. Research [21], [22] confirms that ear canals exhibit subtle yet significant geometric asymmetries, including variations in length, curvature, and cross-sectional area. These asymmetries, along with individual speaking patterns, contribute to the observed pressure differences.

Based on these observations, we hypothesize that the SPS for specific speech content encapsulates unique and repeatable spatiotemporal features for each speaker. As illustrated in Fig. 2(b), SPSs show strong intra-speaker consistency and notable inter-speaker variability. For instance, when two volunteers (A and B) each pronounce "Video my mom" twice, samples from the same speaker demonstrate high similarity, while those from different speakers display significant variation. These findings underscore the potential of BaroAuth in utilizing SPSs for effective speaker authentication.

III. Data Collection

We develop a python application to run on a PC for simultaneous collection of microphone and barometer signals through BLE (Bluetooth Low Energy) communication with an Arduino sensing board, operating at sampling rates of 44,100 Hz for the microphone and 108 Hz for the barometer. We recruit 25 volunteers (four women and twenty-one men), aged between 21 and 50 for the study. As illustrated in Fig. 10(b), each participant wears the Barobuds device and either stands or sits while issuing voice commands, ensuring minimal head movement. The study is conducted in accordance with the ethical standards set by the Science and Technology Ethics Committee of Donghua University. Informed consent is obtained from all participants prior to the experiment. Participants are fully informed of the study's objectives, procedures, and their rights to withdraw at any time without consequence.

Each volunteer preselects *ten* commands from Table I, which lists 30 commands divided into *three* groups by length, ensuring

TABLE I
COMMAND LIST

Command		
2-3 words (*nine* commands)		
1. OK Google.	2. Hey Siri.	3. Take photo.
4. Call Nancy.	5. Turn off Wi-Fi.	6. Video my mom.
7. Check e-mail.	8. Lock screen.	9. Disable all alarms.
10. Open google maps.		
4-5 words (*ten* commands)		
11. Navigate to my office.	12. How's the weather today?	
13. Find my way home.	14. How are you today?	
15. Turn on the lights.	16. Turn off my reminders.	
17. What's on my calendar next?	18. Send a message to mom.	
19. Open up google scholar website.	20. Take me to nearby supermarket.	
6-7 words (*ten* commands)		
21. Where is the closest grocery store?	22. Can you translate this into Chinese?	
23. Can you show me nearby hotels?	24. Play my favorite song by Apple Music.	
25. Who won the final game last night?	26. I think you look really cool today.	
27. Help me find the nearest gas station.	28. What's the current temperature in Shanghai?	
29. I would like to order KFC takeout.	30. Show me around the local tourist attractions.	

an even distribution across groups. Volunteers coordinate to ensure each command is selected $four$ to six times. We then collect $three$ types of data:

Trace A: Collected over 30 days. Each volunteer speaks each command ten times daily with different speeds and vocal amplitude. Specifically, each command is spoken six times with varying speeds (two times at normal, slow, and fast speeds) and twice at normal speed with small and large vocal amplitude. To study the robustness of BaroAuth in motion scenarios, we conduct data collection during the last 15 days in $three$ different mobile scenarios: walking, climbing stairs, and riding an elevator, each for $five$ days. This results in 300 microphone signals and corresponding pressure sensor signals per command per volunteer. We also record the entire process with a video camera for future mimic attacks.

Trace B: For mimic attacks, we select $five$ volunteers as victims, with others as attackers. Each attacker performs ten zero-knowledge attacks (see Scenario A) for each victim's command. Then, after observing the victims' mouth movements while speaking the target command in videos, the attackers attempt to mimic the commands ten times (see Scenario B). Each command undergoes 100 attacks for both scenarios.

Trace C: Collected one week (six months after Trace A). Volunteers speak each command ten times daily at normal speed and articulatory motion amplitude.

IV. Design of BaroAuth

A. Overview

The primary concept behind BaroAuth is to utilize in-ear wearable devices to capture the unique dynamic deformation patterns of the ear canal during speech. When a user speaks a command, both voice and pressure signals from the left and right ear canals are recorded simultaneously using a microphone and pressure sensor. It is crucial to emphasize that the voice data is solely used for preprocessing and pseudo sample generation, with no involvement in the model's training process.

BaroAuth implements a two-phase authentication framework, as illustrated in Fig. 3. The registration and training phase first collects SPSs and speech signals, followed by essential *Data Preprocessing*. The preprocessed data then undergoes *Pseudo Sample Generation* and signal-level augmentation to enhance model robustness. These processed samples are used to train and fine-tune the BaroIDNet with a hybrid loss function. It is noteworthy that the pre-trained BaroIDNet is trained on a shared dataset, collected with participants' consent through the *Data Collection* module (detailed in Section. III).

During the authentication phase, verification samples are preprocessed following the same pipeline before being evaluated by the optimized BaroIDNet, which is efficiently deployed on mobile devices through ONNX[1] runtime for edge inference.

B. Data Preprocessing

In our model, pressure data from both the left and right ear canals undergo the same preprocessing pipeline. For simplicity, the SPSs are treated as pairs in the following sections. When provided with a pair of time-aligned sequences from the microphone and pressure sensor, we apply the subsequent preprocessing steps:

1) Voice Activity Detection: Since our focus is solely on the pressure time series related to human speech activities, WebRTC [23] is used to extract speech segments from the audio signal. Specifically, given an audio signal A, it is segmented into multiple fragments based on voice commands, with the corresponding SPS being divided synchronously and denoted as P. This procedure results in several pairs of microphone and barometer data segments. Each fragment is represented as $D_k = (A_k, P_k), k \geqslant 1$, where A_k and P_k correspond to the audio sequence and the associated SPS, respectively. During segmentation, it is important to note that D_k exhibits a certain degree of lead and lag relative to P_k (as shown in Fig. 2(a). Let $A_k = \{a_{k,t_0}, a_{k,t_1}, \ldots, a_{k,t_T}\}$, where t_0 and t_T represent the start and end time points of A_k, respectively. The complete SPS is then extracted as $P_k = \{p_{k,t_0-\sigma}, \ldots, p_{k,t_0}, \ldots, p_{k,t_T}, \ldots, p_{k,t_T+\sigma}\}$, where the parameter σ is used to adjust the time boundaries.

The WebRTC-VAD-based segmentation above is used *only* in the training stage to form paired fragments $D_k = (A_k, P_k)$ and to create rhythm-transformed pseudo samples (detailed in the subsequent preprocessing pipeline). **During inference**, the authentication pipeline relies solely on the pressure stream, without any audio features. Concretely, we perform a lightweight *pressure-only* segmentation on the continuous pressure signals: (i) baseline removal (e.g., via following IV-B2) followed by mild band-pass filtering (0.5–10 Hz) and amplitude squaring; (ii) short-term energy computation to identify local peaks [24]; (iii) boundary extension from each peak until the energy stabilizes to a local baseline; and (iv) temporal normalization of

[1]Open Neural Network Exchange (ONNX) is an open standard format for representing machine learning models that enables interoperability between different frameworks. https://github.com/onnx/onnx

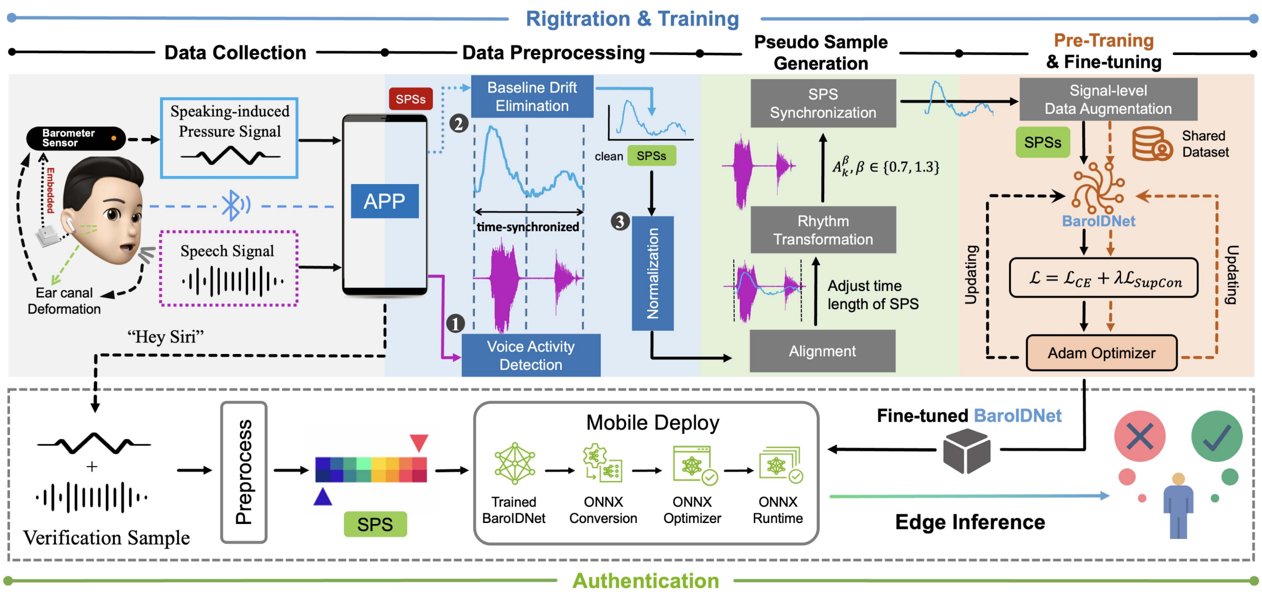


Fig. 3. Overview of BaroAuth.

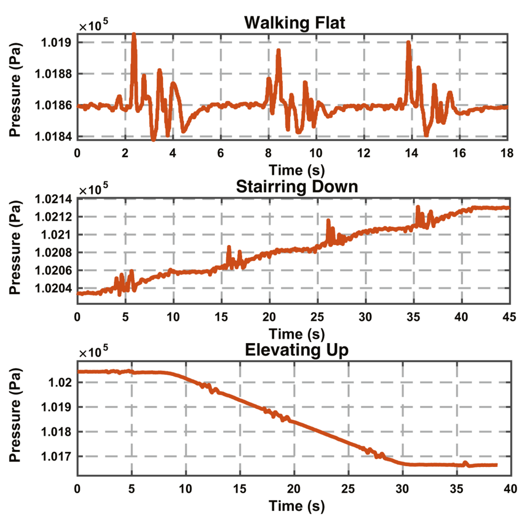


Fig. 4. SPSs under three classical modalities.

each segment for downstream processing. This design keeps the runtime system entirely audio-independent while remaining consistent with the subsequent preprocessing pipeline.

2) Baseline Drift Elimination: Due to the pressure sensor's characteristics, sudden changes in altitude or external temperature can cause baseline drift in the SPSs detected by BaroAuth, leading to varying degrees of distortion in P_k and reducing its accuracy. To ensure practicality, this study focuses on three common mobile scenarios: walking, climbing stairs, and taking an elevator. As illustrated in Fig. 4, while walking on flat ground, the pressure signals remain stable with no baseline drift. Furthermore, minor head movements during speech have no impact on ear canal pressure readings, as the Bosch BMP390 pressure sensor in BaroAuth can detect altitude changes as small as 8 cm. However, during stair climbing and elevator rides, significant baseline drift is observed, which distorts the SPSs. Specifically, ear pressure decreases with an increase in altitude and increases as altitude decreases.

Empirical Mode Decomposition (EMD) [25] is used as a filter bank to eliminate low-frequency baseline drift signals and extract drift-free SPSs. Unlike traditional methods that depend on predefined basis functions (such as sine functions in Fourier transforms or mother wavelets in wavelet transforms), EMD is a data-driven approach that adaptively decomposes the input signal into multiple frequency components based on its intrinsic characteristics. Each of these components represents a distinct oscillatory mode. Specifically, EMD assumes that a complex signal $g(x)$ can be decomposed into x sub-signals, called Intrinsic Mode Functions (IMFs). Each IMF is a zero-mean, narrowband oscillatory signal that captures the oscillatory behavior of the signal across different frequency ranges.

To decompose the SPS with baseline drift $\tilde{P}_k(t)$ into a finite number of IMFs, EMD proceeds as follows:

- Identify all extrema of $\tilde{P}_k(t)$.
- Interpolate between all local minima (or maxima) to obtain the envelopes, $E_{\min}(t)$ and $E_{\max}(t)$.
- Compute the mean $L(t) = [E_{\min}(t) + E_{\max}(t)]/2$.
- Extract the detail signal $C(t) = \tilde{P}_k(t) - L(t)$.
- Iterate on the residual signal $L(t)$ until it no longer satisfies the zero-mean and narrowband oscillation conditions.

Finally, the signal $\tilde{P}_k(t)$ is decomposed as $\tilde{P}_i(t) = \sum_{m=1}^{M} C_m(t) + L_M(t)$, where $C_m(t)$ represents the m^{th} IMF corresponding to different frequency-local oscillations, and $L_M(t)$ denotes the final residual component, which captures the global low-frequency trend (such as baseline drift). By removing the low-frequency residual $L_M(t)$, the target signal $P_k(t)$ with no baseline drift can be extracted.

Fig. 5 illustrates the EMD decomposition results of an ear canal pressure signal recorded by Barobuds while the user is speaking in an elevator. Following the EMD decomposition,

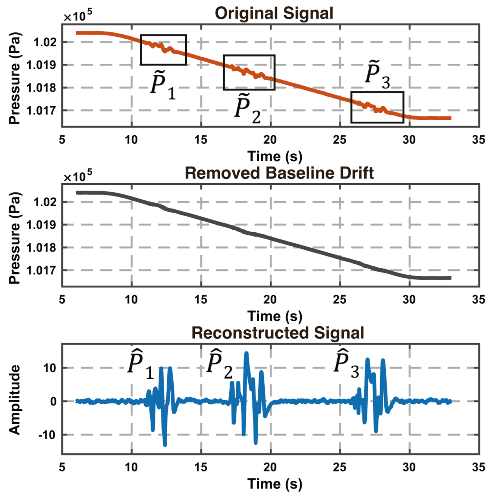


Fig. 5. An example of removed baseline drift and reconstructed SPSs.

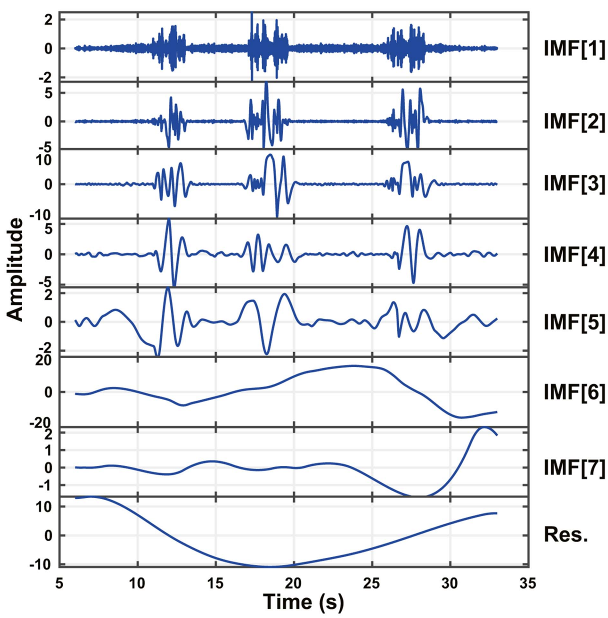


Fig. 6. An example of EMD decomposition.

BaroAuth extracts $seven$ IMFs and one residual. The first IMF corresponds to high-frequency noise from the sensor, while IMF[2-5] capture the frequency components related to the SPS. The remaining IMFs represent low-frequency baseline drift components. As shown in Fig. 6, using IMF[2-5], the drift-free SPS, denoted as $\hat{P}_k(t)$, is reconstructed. To assess the effectiveness of the EMD algorithm, a comparison is made between $\hat{P}_k(t)$ and $P_k(t)$, which was recorded in a stationary state under the assumption that the user speaks the same content with the same rhythm. The comparison results, shown in Fig. 7, reveal a Pearson correlation coefficient [26] of 0.81 between the two signals, indicating a high degree of similarity between $\hat{P}_k(t)$ and $P_k(t)$. Furthermore, EMD naturally eliminates high-frequency sensor noise during the decomposition process.

3) Normalization: Since the barometer directly measures the surrounding atmospheric pressure, the baseline pressure detected by BaroAuth can vary across different sessions. Moreover,

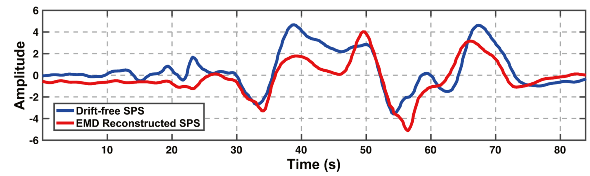


Fig. 7. An example of SPSs decomposed via EMD and collected from a drift-free barometer.

the varying amplitude of articulation motions during speech results in different levels of ear canal deformation, leading to variations in ear canal pressure oscillations. To mitigate these effects, Min-Max normalization is applied to each pressure segment P_k, scaling the values to the range of [-1, 1].

C. Pseudo Sample Generation

The SPSs are significantly influenced by the user's speaking rhythm. Directly using raw data for training reduces the model's adaptability to inputs with varying rhythms, thus weakening its generalization capability. To address this challenge, we propose a rhythm transformation-based pseudo sample generation scheme, which transforms each segment V_i to varying degrees, generating rhythm-diverse P_k and greatly improving the model's robustness to different rhythms. Specifically, the process involves the following steps:

1) Alignment: As discussed in Section IV-B1, the SPS exhibits varying degrees of lead and lag relative to the synchronously sampled audio sequence. To align the pressure signal P_k with A_k on the time axis and enable subsequent rhythm transformations of the SPS, we resample P_i, adjusting its time length to match that of A_k. Specifically, let the length of the pressure signal P_k be n_P and the length of the audio signal A_k be n_A. The following steps are then carried out:

$$t_i^P = \frac{i}{n_P - 1}, i \in [0, n_P - 1], \tag{1}$$

$$t_j^V = \frac{j}{n_A - 1}, j \in [0, n_A - 1], \tag{2}$$

To align the signals, the time points of the audio signal t_j^V are mapped onto the time axis of the pressure signal, resulting in the corresponding positions:

$$t_j^P = \frac{j}{n_A - 1} \times (n_P - 1), \tag{3}$$

Since t_j^P may not align exactly with the sampling points of the pressure signal, its corresponding value is calculated through interpolation. If t_j^P falls between t_i^P and t_{i+1}^P, the pressure signal value p'_{k,t_j} is computed using linear interpolation:

$$p'_{k,t_j} = p_{k,t_i} + \frac{p_{k,t_{i+1}} - p_{k,t_i}}{t_{i+1}^P - t_i^P} \times \left(t_j^P - t_i^P\right), \tag{4}$$

The above process is repeated for all t_j^A values, resulting in the resampled pressure signal:

$$P'_k = \{p'_{k,t_0}, p'_{k,t_1}, \ldots, p'_{k,t_T}\}, \tag{5}$$

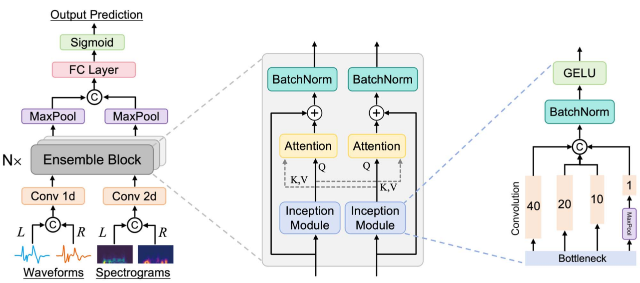


Fig. 8. The architecture of BaroIDNet.

2) Rhythm Transformation: Since rhythm characteristics of audio signals in the time domain (such as syllables and pauses) are more pronounced, rhythm transformations, including time stretching and compression, are applied to A_k to simulate speech patterns at varying speaking speeds. Using a phase vocoder [27], time scaling is performed on A_k while maintaining its pitch. First, the Short-Time Fourier Transform (STFT) is applied to obtain the spectral representation $\mathbb{F}(A_k)$. The time axis of the spectrum is then adjusted using a speed ratio $\beta \in [0.7, 1.3]$ with a step size of 0.1, generating a set of transformed spectra $\{\mathbb{F}(A_k'^{\beta})\}$. Finally, for each β, the corresponding rhythm-transformed audio signals $\{A_k'^{\beta}\}$ are reconstructed using the inverse STFT: $A_k'^{\beta} = \mathbb{F}^{-1}(\mathbb{F}(A_k'^{\beta}))$.

3) SPS Synchronization: Since A_k and P_k' are synchronized on the time axis (after *Alignment*), applying rhythm transformations to the audio signal A_k results in a set of transformed signals $\{A_k'^{\beta} | \beta \in [0.7, 1.3]\}$, each representing a different speaking rhythm. For each $A_k'^{\beta}$, P_k' undergoes the same alignment process to generate $P_k'^{\beta}$. Ultimately, a set of synchronized pairs $D_k' = \{(A_k'^{\beta}, P_k'^{\beta}) | \beta \in [0.7, 1.3]\}$ is formed. As a result, the original D_k is augmented to produce six additional copies with varying speaking rhythms (except $\beta = 1.0$), creating the augmented set D_k'.

D. Authentication Network

1) Design of BaroIDNet: The augmented set D_k' is then input into **BaroIDNet** (as shown in Fig. 8), where time- and frequency-domain features are integrated at the initial stage. Multi-scale convolution and cross-domain attention mechanisms are used to enable deep feature interaction and fusion. To improve training stability, a residual regularization strategy is applied, transforming the raw signals into compact, efficient feature vectors for authentication.

a) Parallel Representation Construction: By analyzing raw time series data from multiple participants (after adjusting for variations in earbud performance), we find that SPSs from the binaural channel exhibit considerable variation across individuals. While some participants show high similarity between the left and right ear canals, others display noticeable differences. This inter-ear canal variability is highly specific to each individual, reflecting subtle personal traits that offer valuable information for identity recognition. To fully leverage this property, the SPSs from the binaural channel are concatenated in both the time and frequency domains, followed by the application of convolutional operations for initial feature extraction.

Let $\mathbf{z}_t^{(L)}, \mathbf{z}_t^{(R)} \in \mathbb{R}^T$ represent the time-domain pressure sequences from the left and right ear canals, both of length T. These signals are transformed into spectrograms $\mathbf{Z}_f^{(L)}, \mathbf{Z}_f^{(R)} \in \mathbb{R}^{F \times T'}$ using STFT, where F denotes the frequency resolution and T' is the number of time steps after framing. The concatenated SPSs from both ear canals are then fed through convolutional layers for feature extraction.

In the time domain, one-dimensional convolution (Conv1D) is used to extract features:

$$\mathbf{H}_t = \text{Conv1D}([\mathbf{z}_t^{(L)}, \mathbf{z}_t^{(R)}]) \in \mathbb{R}^{d \times T}, \tag{6}$$

In the frequency domain, two-dimensional convolution (Conv2D) is applied to the spectrograms, yielding:

$$\mathbf{H}_f = \text{Conv2D}([\mathbf{Z}_f^{(L)}, \mathbf{Z}_f^{(R)}]) \in \mathbb{R}^{d \times T''}, \tag{7}$$

where d is the feature dimension, and T'' represents the time steps after convolution and pooling operations.

These initial feature mappings effectively integrate the time-frequency characteristics of the left and right ears, providing a rich set of foundational features for subsequent multi-scale modeling and cross-domain fusion.

b) Multi-scale Feature Extractor: To capture features across multiple temporal scales in the ear pressure signals, Inception modules [28] are applied to both $\mathbf{H}_t$ and $\mathbf{H}_f$. This allows for multi-scale modeling via parallel branches, each with different convolution kernel sizes, along with a pooling branch. The kernel sizes are dynamically defined as $[k, k/2, k/4]$, where k represents the initial receptive field size. Each convolution branch produces feature maps with a channel size of $d_{\text{model}/}4$, thus balancing model complexity with representation power. For the pooling branch, we first apply 3×3 max pooling (MaxPool), followed by a 1×1 convolution to map the feature dimensions and capture the global context. The outputs from all branches are then integrated, forming the final feature representations as:

$$\mathbf{S} = [\mathbf{s}^k; \mathbf{s}^{k/2}; \mathbf{s}^{k/4}; \mathbf{s}^{\text{pool}}], \tag{8}$$

where $\mathbf{s}^k$ corresponds to the outputs from convolution branches with different kernel sizes, and $\mathbf{s}^{\text{pool}}$ denotes the output of the pooling branch.

Compared to the original Inception design, we introduce the following adaptations tailored to the characteristics of SPSs: i) setting different receptive field sizes for time-domain and frequency-domain features; ii) incorporating Batch Normalization and the GELU activation function after each branch to enhance training stability; and iii) introducing a dynamic adjustment mechanism for the convolutional channel sizes in multiple branches to balance computational cost and representation capacity. This design enables BaroIDNet to effectively capture global trends, local patterns, and short-term dynamic changes across different temporal scales, significantly improving feature representation capability.

c) Feature Fusion: SPSs display intricate characteristics in both time and frequency domains, such as instantaneous dynamic changes and harmonic energy distributions. To effectively combine these complementary features, we propose a cross-attention-based fusion mechanism. This mechanism dynamically aligns and models the interactions between time- and frequency-domain features, ensuring efficient and synergistic optimization of the features.

Given time-domain features $\mathbf{S}_t \in \mathbb{R}^{d_t \times T}$ and frequency-domain features $\mathbf{S}_f \in \mathbb{R}^{d_f \times F}$, the mechanism employs *Query*, *Key*, and *Value* mappings to capture dependencies between the two domains. For instance, when time-domain features attend to frequency-domain features, $\mathbf{S}_t$ is used as the *Query* (Q_t), while $\mathbf{S}_f$ serves as both the *Key* (K_f) and *Value* (V_f). The attention mechanism then computes similarity scores to derive weights, effectively capturing the interactions between temporal and spectral components:

$$\mathrm{Attention}_{t \to f} = \mathrm{Softmax}\left(\frac{\mathrm{Q}_t \mathrm{K}_f^T}{\sqrt{d_k}}\right) \mathrm{V}_f, \tag{9}$$

where $\mathrm{Q}_t = \mathrm{W}_q \mathbf{S}_t$, $\mathrm{K}_f = \mathrm{W}_k \mathbf{S}_f$, and $\mathrm{V}_f = \mathrm{W}_v \mathbf{S}_f$. $\mathrm{W}_q \in \mathbb{R}^{d_k \times d_t}$, $\mathrm{W}_k, \mathrm{W}_v \in \mathbb{R}^{d_k \times d_f}$ are learnable linear projection matrices. These matrices project input features into a unified feature space, with d_k representing the dimension of the projected features.

To improve joint representation learning, the attention output is enhanced through residual connections and normalization, leading to the updated time-domain features $\mathbf{S}'_t$. Similarly, for frequency-domain features attending to time-domain features, cross-attention is computed, resulting in $\mathbf{S}'_f$. This bidirectional attention mechanism effectively captures the complementary information between the two domains. Finally, the fused representation is obtained by concatenating $\mathbf{S}'_t$ and $\mathbf{S}'_f$.

d) Feature Combination and Classification: After the convolution, Inception modules, and cross-attention fusion, a global pooling layer is applied to compress and integrate the time- and frequency-domain features into a unified representation, denoted as $\mathbf{F}$. Global pooling, such as adaptive average pooling, helps reduce the dimensionality while preserving key temporal and frequency information. The resulting feature vector $\mathbf{F}$ captures both the detailed dynamic changes and the global spectral patterns. This vector is then passed through a fully connected layer, which maps it to the classification space to assess user legitimacy.

2) Loss Function Design: Inspired by Supervised Contrastive Learning [29], we attempt to combine decision boundary optimization with feature space structure, further improving the authentication performance of BaroAuth. We design a hybrid loss function below:

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda \cdot \mathcal{L}_{SupCon}, \tag{10}$$

where the weight parameter λ balances the contribution between classification and contrastive objectives, optimized through systematic ablation studies.

The CE (Cross-Entropy) component provides end-to-end verification signals:

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log \sigma(\hat{y}_i) + (1 - y_i)\log(1 - \sigma(\hat{y}_i))], \tag{11}$$

where N denotes the batch size, $\hat{y}_i$ is the predicted output, $\sigma(\cdot)$ is the Sigmoid activation function, and $y_i \in \{0, 1\}$ denotes the ground-truth label.

For the feature space optimization, we implement an enhanced SupCon (Supervised Contrastive) loss function:

$$\mathcal{L}_{SupCon} = \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\mathbf{f}_i \cdot \mathbf{f}_p / \tau)}{\sum_{a \in H(i) \cup \mathcal{M}} \exp(\mathbf{f}_i \cdot \mathbf{f}_a / \tau)}, \tag{12}$$

Here, $\mathbf{f}_i$ represents the L2-normalized feature vector of the anchor sample, $\mathbf{f}_p$ denotes the feature vector of a positive sample, I denotes the set of sample indices in the current batch, $P(i)$ contains positive samples sharing identity with anchor i, τ is the temperature parameter controlling the concentration of feature distribution, while $H(i)$ and $\mathcal{M}$ represent the hard negative samples and memory bank respectively, which we detail below.

a) Hard Negative Mining with Memory Bank: To enhance the discriminative power of feature representations, we implement hard negative mining with a memory bank mechanism. The memory bank $\mathcal{M}$ is implemented using a double-ended queue with capacity C:

$$\mathcal{M} = \{(\mathbf{f}_i, y_i) | i = 1, \ldots, C\}, \tag{13}$$

For each anchor feature $\mathbf{f}_a$ in a batch, we first identify negative samples from the memory bank by label comparison:

$$\mathcal{N}_a = \{n | y_n \neq y_a, (\mathbf{f}_n, y_n) \in \mathcal{M}\}, \tag{14}$$

where y_n and y_a denote the labels of the negative sample and anchor sample respectively, and $(\mathbf{f}_n, y_n)$ represents a feature-label pair stored in the memory bank $\mathcal{M}$.

The similarity between the anchor and negative samples is computed as:

$$S(a, n) = \mathbf{f}_a \cdot \mathbf{f}_n^T, \tag{15}$$

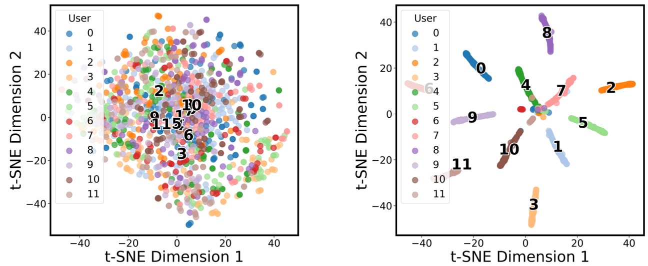
where $\mathbf{f}_a$ denotes anchor feature and $\mathbf{f}_n$ denotes negative sample feature. Since both features are L2-normalized, this dot product effectively computes their cosine similarity.

We select the top-k most similar negative samples as hard negatives $H(i)$. Their importance weights are computed using temperature-scaled softmax:

$$w_n = \frac{\exp(S(a, n)/\tau)}{\sum_{n' \in \text{top-k}} \exp(S(a, n')/\tau)}, \tag{16}$$

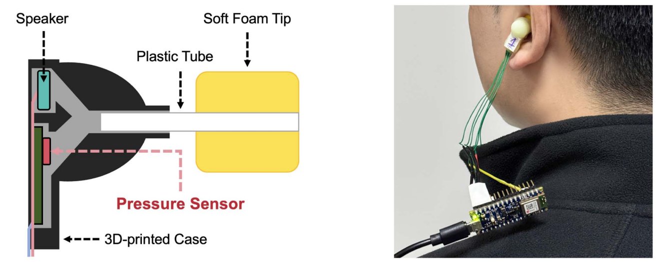
where τ is the same temperature parameter as used in the SupCon loss. The memory bank is updated after each batch by appending new features and their labels:

$$\mathcal{M} \leftarrow (\mathcal{M} \setminus \{\text{oldest } |\text{batch}| \text{ entries}\}) \cup \{(\mathbf{f}_i, y_i)\}_{i=1}^{|\text{batch}|}, \tag{17}$$



(a) Raw Data (b) Fusion Features

Fig. 9. The t-SNE visualization of features extracted by raw data and BaroIDNet. (a) Raw Data. (b) Fusion Features.



(a) Hardware Schematic (b) Experimental Setup

Fig. 10. Barobuds design showing (a) internal hardware components and (b) practical wearing configuration. (a) Hardware Schematic. (b) Experimental Setup.

This mechanism enables our model to leverage historical feature information and focus on the most challenging negative samples during training, effectively improving the model's discriminative capability.

Finally, to assess BaroAuth's capability in extracting user-specific features, we use t-SNE [30] to visualize both the raw input data and the features extracted by the model. In Fig. 9(a), the t-SNE plot of the raw input data shows that samples from different users are intermixed, suggesting a lack of clear discriminative properties. In contrast, Fig. 9(b) presents the features extracted by the model, where samples from the same user are grouped into distinct, compact clusters. These results highlight BaroAuth's ability to transform ambiguous raw inputs into a well-structured feature space, where user-specific separability is evident, demonstrating its potential for identity verification.

E. Shared Backbone for Enrollment

We additionally preserve a shared backbone during training to enable efficient new-user enrollment. The backbone consists of the feature extraction network, including residual convolutional blocks, the cross-attention module, and the embedding layer, while excluding the final user-specific classifier. Batch normalization affine parameters are retained, whereas the running statistics are later recalibrated. At the beginning of training, the backbone parameters are copied into a global exponential moving average (EMA) model, denoted as θ_{ema}. The EMA coefficient α is set within $[0.05, 0.2]$ (0.1 by default), and is linearly warmed up from 0 to 0.1 during the first two epochs to mitigate instability in early training. During user training, each model is optimized independently following the standard one-vs-all protocol. At the end of every epoch, the corresponding backbone is aggregated into θ_{ema} using:

$$\theta_{\text{ema}} \leftarrow (1-\alpha)\cdot\theta_{\text{ema}} + \alpha\cdot\theta^{(u)}_{\text{backbone}}. \quad (18)$$

To account for imbalanced user data sizes, the effective α is scaled according to the number of samples, clipped to remain within $[0.05, 0.2]$. Upon completion, two outputs are obtained: the set of private user-specific models and one shared backbone (*SharedBackbone_EMA*). To avoid biased batch normalization statistics from individual users, we recalibrate the shared backbone using a small calibration set that mixes samples from multiple users. Specifically, the model is set to training mode and forward-propagates several batches without backpropagation, thereby updating the BN running means and variances, before being switched back to evaluation mode with fixed statistics. This shared backbone can then serve as a universal initialization for new-user enrollment.

V. Implementation

Hardware: BaroAuth comprises a custom-designed pair of earbuds, an Arduino sensing board, a Windows laptop, and a Linux server. As depicted in Fig. 10(a), we develop a prototype of the earbuds using a 3D-printed case, adapting design parameters from OpenEarable [31]. The PCB circuit board is modified to integrate a Bosch BMP390 barometric pressure sensor. A NANO 33 BLE Sense development board is employed to acquire and transmit barometer signals at a sampling rate of 108 Hz. For real-time signal monitoring and initial preprocessing, we use a Windows laptop equipped with an Intel Core i7-8750H processor, 16 GB of RAM, and an NVIDIA RTX 1060 GPU. The Linux server, running Ubuntu 18.04, is outfitted with dual AMD EPYC 7542 32-core processors, 64 GB of RAM, and an NVIDIA Tesla V100 GPU, and is responsible for executing BaroIDNet for authentication.

System Parameter Settings: Our BaroAuth system employs a 3-layer residual network processing both time and frequency domain barometric signals. The model combines Cross-Entropy and Supervised Contrastive losses (weight=0.3, temperature=0.2) with hard negative mining and a memory bank (size=128). During training stage, we use Adam optimizer (lr=0.0001, batch size=128) for 50 epochs with data augmentation (probability=0.7) including time shifts, amplitude scaling, and noise injection. Barometric signals are sampled at 108 Hz in 2-second windows, generating 64-dimensional embeddings. For evaluation, we adapt a one-vs-all verification protocol across 25 enrolled users as the primary benchmark, and further assess practical few-shot enrollment scenarios in Section IV-E to emulate real-world new-user registration.

Model Training and Testing: BaroIDNet is implemented in PyTorch with training conducted on NVIDIA GPUs. To improve robustness against sensor noise and natural variations, we apply lightweight signal-level augmentations in addition to the rhythm-based pseudo sample generation. These include temporal perturbations (random time shifting, amplitude scaling, Gaussian noise injection) and simple spectral modifications (random masking and band dropout after STFT). All augmentations are applied stochastically during training without altering the underlying rhythm structure. We employ adaptive early stopping with decreasing patience levels [20,15,10,5], StepLR scheduling, and dynamic batch normalization.

Dataset Partitioning and Negative-Sample Selection: We organize the corpus into *(user, command)* groups and perform an utterance-level (per recorded command instance) partition based on a random permutation parameterized by a random seed. Specifically, we first apply a random permutation of all utterances parameterized by a given seed. Then, for each group, we assign the first three utterances to the validation set, the next three to the test set, and all remaining utterances to the training set, ensuring that each utterance appears in exactly one of the three disjoint subsets with no overlap or leakage. Aggregated over all groups, such assignment yields an *empirical* utterance-level ratio of approximately 70%/15%/15% for train/validation/test across the corpus.

For user-dependent verification, we train one binary classifier per target user. In each subset (train/validation/test), *all* utterances from the target user are treated as positives, while *all* utterances from the remaining 24 users in the *same* subset serve as negatives (i.e., an all-impostor policy), with no sub-sampling or class re-balancing. This keeps classes imbalanced by design but yields a conservative estimate of FAR. We further assess robustness via K-fold cross-validation under the same stratification (see Appendix H for details).

Mobile Deployment: For deployment evaluation, ONNX exported models are benchmarked across multiple mobile inference backends (Core ML, HiAI, NNAPI, CPU), measuring data transmission, preprocessing, and inference times separately to assess real-world performance on resource-constrained devices.

VI. Experimental Results and Analysis

A. Baselines and Metrics

Baselines: To thoroughly evaluate the performance of BaroIDNet, we compare it with baseline methods, including SVM (Support Vector Machine), KNN (K-Nearest Neighbors), CNN-LSTM (CNN-Long Short-Term Memory), and TCN (Temporal Convolutional Network). SVM and KNN are classical machine learning models relying on handcrafted features, while CNN-LSTM and TCN are deep learning architectures tailored for time series modeling. This comparison highlights the superiority and robustness of BaroIDNet in processing ear pressure signals. For fairness, all methods use the same dataset splits, training settings, and evaluation metrics.

Metrics: We evaluate BaroAuth using a set of complementary metrics. The False Acceptance Rate (FAR) measures the proportion of illegitimate inputs incorrectly accepted, reflecting system security. Conversely, the False Rejection Rate (FRR) quantifies the proportion of legitimate inputs incorrectly rejected, indicating usability impact. The Equal Error Rate (EER), where FAR equals FRR, provides a balanced single-value performance comparison across systems. Accuracy measures overall correctness, while F1-Score balances precision and recall—critical under

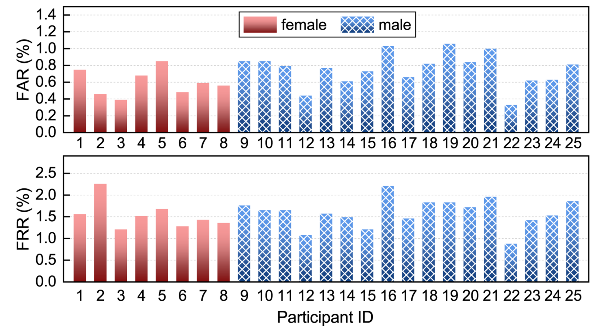


Fig. 11. Overall FARs and FRRs of all participants.

TABLE II
Performance Comparison Across Different Baselines on Monaural and Binaural Channel

Methods	Monaural Channel			Binaural Channel		
	EER	FAR	FRR	EER	FAR	FRR
SVM	26.04%	22.82%	8.05%	25.31%	22.91%	8.15%
KNN	10.33%	8.63%	7.53%	10.43%	8.79%	7.79%
CNN-LSTM	47.11%	45.10%	52.28%	47.21%	45.25%	53.06%
TCN	6.82%	5.13%	4.86%	6.03%	5.23%	5.10%
BaroIDNet	**1.38%**	**1.21%**	**2.45%**	**1.02%**	**0.41%**	**1.23%**

class imbalance in biometric authentication. All metrics are derived from the ROC curve; optimal thresholds are selected by maximizing Youden's J statistic [32].

B. Overall Performance

We first evaluate the overall performance of BaroIDNet based on the collected traces by examining the FAR and FRR metrics. During the experiment, we consider each participant as a legitimate user while treating the remaining participants as attackers, simulating zero-knowledge attacks where attackers attempt to fool the system with their SPSs. As shown in Fig. 11, BaroIDNet achieves consistent performance across all participants, with an average FAR of 0.41% and FRR of 1.23% in the binaural channel setting. To further examine participant-wise variability, We further analyze demographic subgroups. Across genders, both male (15 participants) and female (10 participants) users achieved similar average error rates, showing no significant bias. Age groups (21–27 vs. 28–35 years) also displayed comparable results, suggesting that BaroIDNet generalizes well across young adult populations. Nevertheless, we observed moderate inter-individual variability: the best-performing user attained an FRR close to 0.5% with FAR below 0.3%, while the relatively worst-performing user showed an FRR of about 2.3% and FAR near 1.0%. This discrepancy likely stems from differences in ear-canal morphology and wearing stability. Importantly, even the least favorable case remained within acceptable error rates, underscoring the robustness of BaroIDNet across diverse individuals.

Table II summarizes the performance comparison between BaroIDNet and four baseline methods under both monaural and binaural configurations. BaroIDNet achieves the lowest error rates across the board, with an EER of 1.38% for monaural input

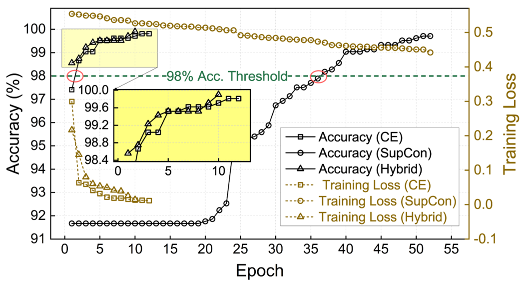


Fig. 12. Accuracy and loss curves under different loss functions.

TABLE III
MODEL PERFORMANCE COMPARISON UNDER DIFFERENT TRAINING STRATEGIES

Method	**Acc.**	**F1**	**EER**	**FAR**	**FRR**	**Train (s)**	**Mem (MB)**
Base	98.96	98.99	1.28	0.88	1.87	200.83	2899.78
HM	99.21	99.22	1.34	0.64	1.61	202.06	**2892.14**
Aug	99.25	99.27	1.21	0.58	1.31	195.76	2900.79
Mega	**99.38**	**99.38**	**1.02**	**0.41**	**1.23**	**183.51**	2920.89

Base: training BaroIDNet with hybrid loss, no strategy.
HM: hybrid loss with the "Hard Negative Mining" strategy.
Aug: hybrid loss with the "Data Augmentation" strategy.
Mega: both *HM* and *Aug* strategies enabled.

and a further reduction to 1.02% in the binaural setting. This improvement highlights the model's ability to leverage complementary information from dual-channel SPS signals through effective feature fusion. In contrast, traditional approaches such as SVM (25.31% EER) and CNN-LSTM (47.21% EER) fail to model the intricate temporal and spectral patterns inherent in SPS data, resulting in substantially degraded performance.

C. Ablation Study

1) Effectiveness of Hybrid Loss: To enhance the discriminative capability of BaroAuth, we evaluate different loss functions, including standard Cross-Entropy (CE) loss, Supervised Contrastive (SupCon) loss, and the proposed Hybrid loss. As shown in Fig. 12, the Hybrid loss rapidly achieves 98.5% accuracy within the first epoch, whereas CE loss requires 2–5 epochs to reach a similar level. SupCon loss demonstrates a markedly different pattern, with accuracy plateauing at 91–92% for nearly 30 epochs before abruptly rising to 98%. Training loss curves further highlight these differences: CE and Hybrid losses converge quickly and stabilize, while SupCon loss continues a gradual descent without a clear convergence point. Overall, the Hybrid loss combines the fast convergence of CE with the stability benefits of contrastive learning, resulting in more efficient and robust optimization.

2) Effectiveness of Training Strategies: We evaluate the impact of different training strategies on BaroIDNet, including the baseline setup (Base), hard negative mining (HM), data augmentation (Aug), and their combination (Mega), as shown in Table III. The Mega strategy delivers the best overall performance, achieving 99.38% accuracy, 0.41% FAR (53.4% lower than Base), and 1.02% EER (20.3% improvement). HM enhances boundary learning by emphasizing difficult samples, while Aug promotes generalization through pattern diversity.

TABLE IV
PERFORMANCE W/ AND W/O ATTENTION MODULES AND UNDER VARYING NUMBERS OF INCEPTION BLOCKS

Metrics	**Attention Module**			**Inception Blocks**			
	Time	Frequency	Time+Freq.	1	3	6	9
FAR	2.23%	3.75%	**1.57%**	2.12%	**1.57%**	1.57%	1.58%
FRR	2.65%	3.16%	**1.31%**	2.13%	**1.31%**	1.30%	1.30%

Their combination improves both intra-class compactness and inter-class separability. In terms of efficiency, Mega also reduces training time to 183.51 s (8.6% faster) and maintains competitive memory usage (2920.89 MB). These results validate the effectiveness of integrating decision-boundary refinement and feature-space expansion, providing a robust and resource-efficient training strategy for mobile biometric authentication systems.

3) Effectiveness of Initial Convolution: As described in the Overall Performance section, BaroIDNet uses Conv1D to extract features from concatenated SPSs and Conv2D to process time-frequency representations, capturing complementary features between binaural signals and enhancing both time-domain and frequency-domain expressiveness. Table II shows that compared to single-ear channels, BaroIDNet reduces FAR and FRR by about 2% in binaural channel. This demonstrates that convolution not only extracts subtle differences and complementary patterns in binaural signals but also integrates time-frequency information, improving feature discrimination and overall system performance.

4) Effectiveness of Attention Module: We evaluate the impact of the cross-attention mechanism on the fusion of SPS time-domain and frequency-domain features. As shown in Table IV, without cross-attention, FAR and FRR are 2.23% and 2.65% for time-domain features and 3.75% and 3.16% for frequency-domain features. This indicates that frequency-domain features are more noise-sensitive and perform worse in isolation. With cross-attention, the model captures the correlation between time-domain and frequency-domain features, leveraging their complementary information. FAR and FRR drop to 1.57% and 1.31%, the best results. This demonstrates that cross-attention dynamically focuses on key regions and effectively integrates temporal dynamics with frequency distributions, significantly improving accuracy and robustness.

5) Effectiveness of Inception Blocks: As shown in Table IV, using only one Inception block results in 2.12% FAR and 2.13% FR, respectively, indicating limited single-scale feature extraction. With $three$ Inception blocks, FAR and FRR drop to 1.57% and 1.31%. However, the number of Inception blocks increase to six and $nine$, performance gains plateau, with FAR and FRR stabilizing around 1.57%-1.58% and 1.30%-1.31%. This suggests that feature extraction saturates, and additional Inception blocks increase redundancy without meaningful improvements. Three Inception blocks balance feature extraction and computational complexity, achieving optimal performance.

D. Impact of Sampling Rate

BaroAuth samples SPSs at 108 Hz by default and achieves an EER of 1.07%. As shown in Fig. 13, performance degrades

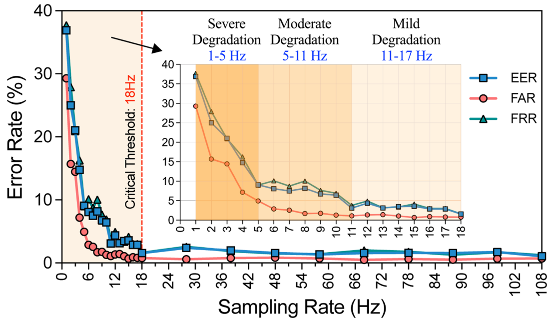


Fig. 13. Error rates with different sampling rates.

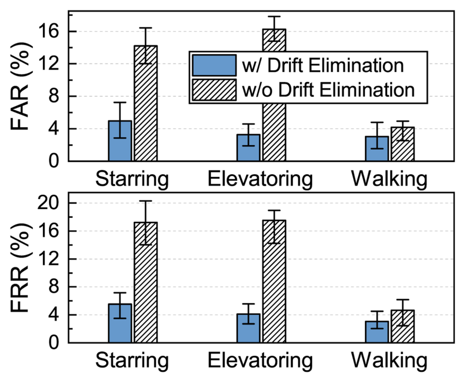


Fig. 14. FARs and FRRs under different mobile scenarios.

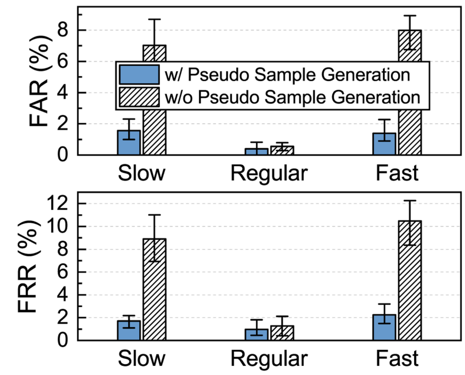


Fig. 15. FARs and FRRs under different speaking rhythms.

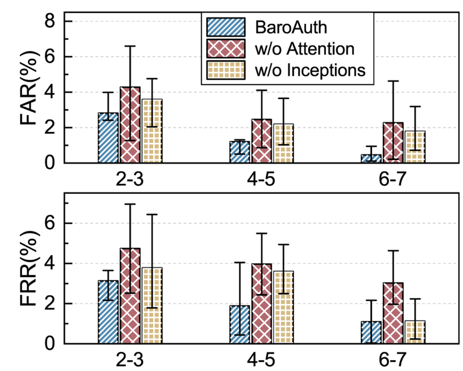


Fig. 16. FARs and FRRs under length of voice commands.

quickly when the sampling rate falls below 5 Hz, becomes acceptable above 11 Hz, and stabilizes around 18 Hz. At 18 Hz, EER, FAR, and FRR all stay below 3% with much lower computation and power cost, indicating that BaroAuth can operate reliably at low sampling rates suitable for mobile devices.

E. Parameter Sensitivity Analysis

We also perform a sensitivity study on key hyperparameters of BaroIDNet, including the memory bank size σ, augmentation probability ω, contrastive loss weight λ, and temperature τ. The results, summarized in Appendix J, show that a configuration of $\sigma = 128$, $\omega = 0.2$, $\lambda = 0.6$ and $\tau = 0.2$ gives near-optimal EER/FAR/FRR, and that performance degrades only moderately when these parameters vary within a reasonable range. This suggests that BaroAuth is not overly sensitive to precise hyperparameter tuning.

F. Robustness Analysis

1) Mobile Scenarios: We evaluate BaroAuth's performance in three mobile scenarios (walking, climbing stairs, and taking an elevator) with and without baseline drift elimination. As shown in Fig. 14, drift elimination significantly improves authentication accuracy across all scenarios. Without drift elimination, FAR and FRR increase notably in "climbing stairs" (16.54%, 17.63%) and "taking an elevator" (14.69%, 15.66%) scenarios. After applying drift elimination, these values decrease substantially to 5.31% and 5.63% for climbing stairs, and 3.22% and 3.66% for taking an elevator. Even in the walking scenario with minimal altitude changes, drift elimination maintains stable performance (FAR: 3.87%, FRR: 3.91%) compared to without (FAR: 3.88%, FRR: 4.32%).

2) Speaking Rhythm: We analyze the effect of rhythm-based pseudo sample generation on the model's robustness to different speaking rhythms. As shown in Fig. 15, without pseudo sample generation, both FAR and FRR exhibit significant fluctuations under non-regular rhythms. FAR increases from 0.45% (regular rhythm) to 7.38% and 7.97% for slow and fast rhythms, respectively. Similarly, FRR rises from 1.25% to 9.57% and 9.76%. With rhythm-based pseudo sample generation, the model demonstrates improved stability. FAR remains consistently low at 1.44%, 0.41%, and 1.93% for slow, regular, and fast rhythms, while FRR stabilizes at 1.52%, 1.23%, and 1.86%, respectively. These results demonstrate that the proposed rhythm-based pseudo sample generation effectively enhances the model's adaptability to varying speaking rhythms.

3) Voice Command Length: We investigate the effect of voice command length on authentication performance. As shown in Fig. 16, longer commands consistently improve system accuracy. Specifically, when the length increases from 2–3 words to 4–5 words, FAR decreases from 2.20% to 1.04%, and FRR drops from 3.02% to 1.92%. The best results are achieved with 6–7 word commands, where FAR and FRR reach 0.41% and 1.23%, respectively. These results indicate a clear positive correlation

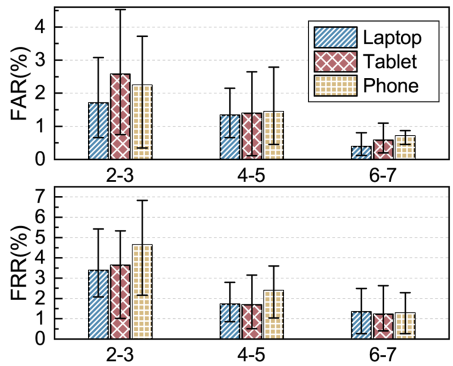


Fig. 17. FARs and FRRs under long-term.

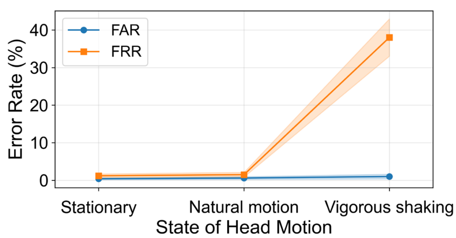


Fig. 18. FARs and FRRs under various head movements.

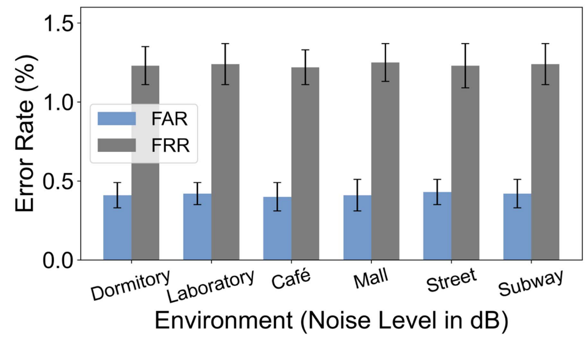


Fig. 19. FARs and FRRs under noisy environments.

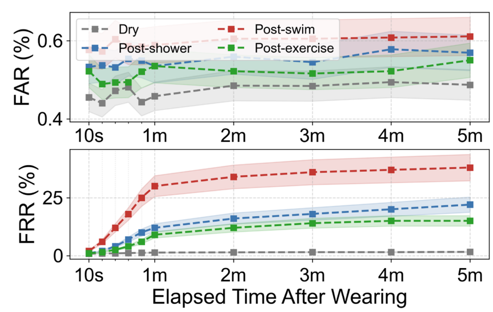


Fig. 20. FARs and FRRs under different ear canal conditions.

between command length and authentication reliability. Longer voice inputs provide richer temporal and semantic patterns, enabling more robust user identification while reducing error rates. This suggests that using moderately long commands (e.g., 6–7 words) can significantly enhance system security and usability in practical deployments.

4) Temporal Stability: We evaluate BaroAuth's long-term authentication performance across different devices and command lengths. As shown in Fig. 17, the system maintains stable performance across all scenarios. For 2-3 word commands, FAR ranges from 2.36% (laptop) to 2.69% (phone), while FRR varies between 3.61% (laptop) and 4.03% (phone). Performance improves with longer commands: for 4-5 words, FAR and FRR decrease to approximately 1.1-1.2% and 1.8-2.2% respectively across devices. The best performance is achieved with 6-7 word commands, where FAR remains below 0.6% and FRR below 1.5% for all devices, demonstrating BaroAuth's robust temporal stability.

5) Head Movements: We evaluate BaroAuth under three head-movement conditions: (i) *stationary*, where participants keep their head still while seated; (ii) *natural motion*, involving light nodding or head turns at $\sim$0.5–1 Hz ($\approx$5–10 times in 10 seconds) with small angles of 10–15°; and (iii) *vigorous shaking*, where participants deliberately move their heads left–right/up–down at $\sim$1–2 Hz ($\approx$10–20 times) with larger angles of 30–45°, serving as a stress test for seal stability. Each condition was repeated across 10 participants with 5 trials each. As shown in Fig. 18, performance remains stable from the stationary baseline (FAR $= 0.4\% \pm 0.2$, FRR $= 1.2\% \pm 0.4$) to natural motion (FAR $= 0.6\% \pm 0.3$, FRR $= 1.5\% \pm 0.5$), while vigorous shaking drastically increases FRR to $38.0\% \pm 4.0$ due to earbud loosening and seal loss.

6) Noisy Environments: Conventional voice-based authentication is vulnerable to ambient noise, leading to severe performance degradation. In contrast, BaroAuth is immune to this issue as it relies solely on ear-canal pressure sequences. To validate this property, we evaluate six representative noise environments: dormitory (38 dB), laboratory (45 dB), café (55 dB), mall (65 dB), street (70 dB), and subway (75 dB). All models are trained under quiet conditions and tested with background noise played through speakers at 40 cm distance. As shown in Fig. 19, BaroAuth maintains consistently low error rates across all noise levels, confirming its robustness to acoustic interference.

7) Moisture Conditions: In real-world use, residual moisture after showering, swimming, or exercising can weaken the in-ear seal. To evaluate robustness, we recruited five participants and tested BaroAuth under four conditions: *dry (air-conditioned)*, *post-shower*, *post-swim*, and *post-exercise*. Participants wore the earbuds immediately without drying and performed repeated authentications for five minutes. As shown in Fig. 20, FAR remained stable in all conditions (0.4% to 0.6%). FRR, however, showed different degradation patterns depending on the scenario: in the *post-swim* case it increased from 6% at 1 min to 38% at 5 min, in the *post-shower* case it reached 22% at 5 min, and in the *post-exercise* case it rose more gradually to 15%. In contrast, the *dry* case consistently stayed below 2%. Once the ear canal dried, FRR quickly recovered to below 2% while FAR remained between 0.5% and 0.6%. These findings indicate that the observed performance decline is mainly caused by humidity-induced seal loosening, which is a common challenge for in-ear earbuds.

TABLE V
COMPUTATIONAL OVERHEAD ON VARIOUS SMART DEVICES

Devices	Release Year	Inference Backend	Time Consumption (ms)				**Total (ms)**
			Data Transmission	Data Preprocessing		Inference	
				VAD	Drift Elimination		
iPhone 15 Pro	2023	Core ML	272.11	8.83	0.43	36.68	318.05
iPhone14 Pro Max	2022	Core ML	264.78	8.46	0.45	40.45	314.14
iPhone11	2019	Core ML	285.75	10.01	1.03	81.63	378.42
Huawei Mate 60 Pro	2023	HiAI	278.92	9.12	0.49	66.05	354.58
Huawei Nova 8	2020	HiAI	296.14	10.30	1.09	95.63	**403.16**
Huawei Mate 20	2018	HiAI	298.93	10.81	1.13	58.35	369.22
Vivo Iqoo Pro 6	2019	NNAPI	286.38	10.08	1.09	55.13	352.68
Xiaomi 15	2025	NNAPI	264.26	8.49	0.42	33.67	306.84
iPad Pro	2018	Core ML	283.43	9.13	0.47	36.85	329.88
iPad 8	2020	Core ML	267.96	11.75	1.55	100.91	382.17
Macbook Pro M2 Pro	2023	CPU	194.22	8.44	0.42	17.34	**220.42**
Lenovo Legion Y7000P	2018	CPU	263.77	8.21	0.43	30.31	302.72

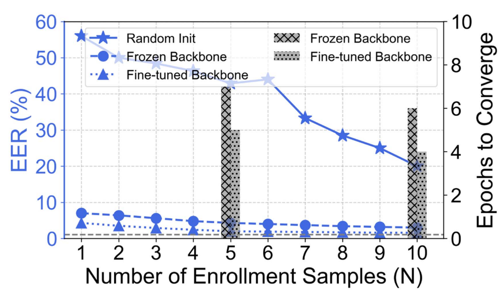


Fig. 21. EERs with varying numbers of enrollment samples from new user.

G. Few-Shot Enrollment

A practical authentication system must support efficient enrollment for new users. In BaroAuth, each user maintains a private one-vs-all model, while the shared backbone described in Section IV-E provides a reusable initialization for rapid onboarding. We adopt a leave-one-user-out (LOUO) protocol: in each round, one user's data is excluded from backbone training and used as enrollment, contributing $N \in \{1, \ldots, 10\}$ utterances (4–5 words), while negatives are drawn only from other users to avoid leakage. We compare three initialization strategies: (i) training BaroIDNet from scratch; (ii) freezing the *Shared-Backbone_EMA* and training only the classifier; and (iii) lightly fine-tuning the last residual block and cross-attention module. As shown in Fig. 21, the *Fine-tuned Backbone* delivers the best few-shot performance: EER decreases from 2.9% ($N = 3$) to 2.0% ($N = 5$) and 1.6% ($N = 10$), close to the full-data baseline (1.07%). The *Frozen Backbone* gives 5.6%/4.3%/3.0% under the same conditions, while *Random Init* remains much higher at 48.4%/42.9%/20.0%. In terms of convergence, fine-tuning requires only 5/4 epochs at $N = 5/10$, freezing takes 7/6, whereas Random Init fails to converge quickly, often exceeding 10 epochs and plateauing at high EER. This translates to a practical enrollment cost of only 15–30 seconds of data collection, and few-shot model update can be completed within a few seconds since only a small subset of parameters are fine-tuned.

H. Computational Overhead

To comprehensively evaluate our system's computational efficiency, we conduct experiments across 13 representative smart devices spanning smartphones, tablets, and laptops (released between 2018 to 2025). As shown in Table V, BaroIDNet achieves practical authentication latency across all tested platforms, with total processing times ranging from 220.42 ms (Macbook Pro M2 Pro) to 403.16 ms (Huawei Nova 8). Modern devices demonstrate superior performance, with the Xiaomi 15 and iPhone 15 Pro achieving total overheads of 306.84 ms and 318.05 ms respectively. Our analysis reveals that data transmission consistently dominates the processing pipeline (63-78% of total time), while preprocessing and inference stages contribute minimally on modern hardware (typically <50 ms combined). Performance correlates strongly with device release years, as evidenced by the 55% inference time reduction between iPhone 11 (81.63 ms) and iPhone 15 Pro (36.68 ms). Despite hardware differences, the system maintains consistent performance across different inference backends (Core ML, HiAI, NNAPI, CPU), confirming our optimization strategy's effectiveness and the system's suitability for immediate real-world deployment.

TABLE VI
PER-BUD POWER AND ESTIMATED BATTERY LIFE UNDER DIFFERENT SAMPLING RATES (SUPPLY VOLTAGE = 3.7 V)

SR (Hz)	Current (mA)	Power (mW)	Battery (h), avg	Range (h)
108	5.6	20.72	9.83	5.9–13.4
50	4.7	17.39	11.72	7.0–16.0
18	3.7	13.69	14.88	8.9–20.3

I. Power Consumption.

On the earbud side, BaroAuth adds overhead from continuous sensing and bluetooth transmission. Profiling three sampling rates (108, 50, and 18 Hz) shows per-bud power of 20.72, 17.39, and 13.69 mW, corresponding to estimated battery lives of 9.8, 11.7, and 14.9 h, respectively (Table VI). Considering the average TWS battery capacity (55.07 mAh @ 3.7 V, ranging 33–75 mAh; see Table I in Supplementary), this endurance is comparable to mainstream products (e.g., AirPods Pro 2, Galaxy Buds), indicating practical deployability. On the smartphone side, we conduct 2-hour continuous tests on representative Android devices (in Table V) with background services disabled. As shown in Fig. 22, battery drop ranged from 8.5% (Xiaomi 15) to 12.0% (Huawei Nova 8). Across the test, devices completed 17.9 k–23.5 k authentications, corresponding to average system-level energy costs of 69–218 mJ per authentication, covering transmission, VAD, drift elimination, and inference. These values represent an upper bound under continuous stress testing; in

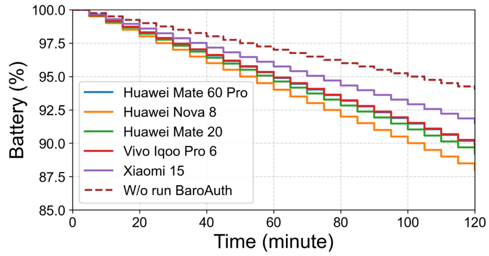
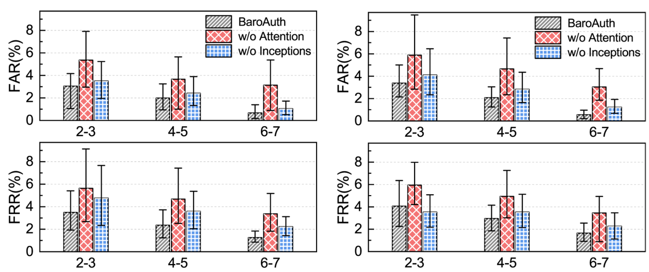


Fig. 22. Battery discharge curves of representative Android smartphones when running BaroAuth authentication service continuously for 2 hours.



(a) Zero-knowledge attack (b) Knowledge-based attack

Fig. 23. Robustness under spoofing attacks. (a) Zero-knowledge attack. (b) Knowledge-based attack.

real usage where authentication is sporadic, the practical energy cost is even lower.

J. Security Evaluation

We evaluate BaroAuth's security robustness through two attack scenarios using Trace B, while also conducting ablation studies to assess the contribution of key architectural components.

1) Zero-Knowledge Attacks: Fig. 23(a) presents BaroAuth's performance under zero-knowledge attacks, revealing strong resilience across all command lengths. Compared to normal usage, the average FAR increases slightly by 0.74% and 0.78% for 2–3 and 4–5 word commands, respectively, while FRR increases by 0.45% and 0.49%. For 6–7 word commands, both FAR and FRR remain unchanged, demonstrating near-complete resistance. These results show that even under adversarial conditions with no prior knowledge, BaroAuth maintains stable and reliable performance. While short commands incur minor error increases, longer commands significantly enhance robustness, reinforcing their value in security-critical applications.

2) Knowledge-Based Attacks: As shown in Fig. 23(b), BaroAuth maintains strong robustness even under more challenging knowledge-based attacks. Compared to normal usage, the average FAR increases by 0.88% and 0.87%, and the FRR increases by 0.67% and 0.75% for 2–3 and 4–5 word commands, respectively. For 6–7 word commands, the increases are minimal—only 0.11% for FAR and 0.21% for FRR—demonstrating near-complete resilience. These results confirm that BaroAuth effectively resists knowledge-based threats, particularly when using longer voice commands. The system's

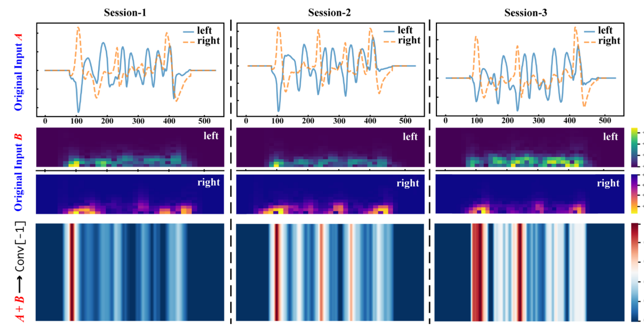


Fig. 24. Model interpretability analysis through Grad-CAM: temporal signals (top), spectrograms (middle), and attention heatmaps (bottom) across three independent sessions from the same subject.

ability to maintain low error rates under adversarial conditions highlights its suitability for high-security voice authentication applications.

K. Model Interpretability

Deep learning models are often regarded as closed boxes, making their decision processes difficult to interpret. To enhance interpretability, we employ Grad-CAM [33], which provides superior localization to reveal input regions most influential to BaroIDNet's classification. Fig. 24 presents attention patterns for a participant's SPSs across three sessions: raw time-domain signals (top), STFT spectrograms (middle), and Grad-CAM heatmaps (bottom). The red regions highlight features with maximum contribution, consistently aligning with vowel segments where vocal tract adjustments generate distinctive ear-canal pressure patterns. Consonant segments, with shorter duration and weaker variations, show limited activation. Spectrograms corroborate this observation, as vowel-dominant regions exhibit higher energy across frequency bands. The cross-session consistency demonstrates the model's ability to capture invariant acoustic signatures unique to each user. This systematic focus on physiologically meaningful features, rather than noise, explains BaroIDNet's superior performance. By jointly leveraging temporal dynamics from raw signals and spectral characteristics from spectrograms, the model achieves robust authentication stable across sessions and environments.

VII. Related Work

A. Earable-Based Sensing Applications

Driven by the miniaturization of MEMS sensors and advancements in edge computing, earables have transformed from simple audio accessories into multifunctional sensing platforms. A recent comprehensive survey by Hu et al. [34] highlights that modern earables leverage diverse sensing modalities, including acoustic, inertial, and physiological sensors, to support a wide range of applications beyond audio playback.

Physiological and Health Monitoring: Earables provide a unique vantage point for capturing vital signs due to their proximity to the temporal artery and respiratory tract.

Recent works have demonstrated continuous heart rate monitoring using audioplethysmography (APG) [35], respiratory rate tracking via body-conducted sounds [36], and even ear pathology screening by analyzing acoustic echoes inside the ear canal [37].

Activity and Behavior Recognition: Integrated IMUs and microphones enable fine-grained activity tracking. For instance, researchers have developed systems to monitor eating behaviors by detecting chewing dynamics [38], track toothbrushing activities via bone-conducted acoustic signals [39], and analyze gait patterns for holistic health assessment [40].

Human-Computer Interaction: Beyond monitoring, earables serve as intuitive interfaces. Innovations in this domain include tracking facial expressions using active acoustic sensing [41] and recognizing silent speech commands by detecting ear canal deformations [42].

While these advancements illustrate the immense sensing potential of earables, the collection of such sensitive physiological and behavioral data raises critical security concerns. This necessitates robust user authentication mechanisms, which we discuss below.

B. User Authentication on Earables

Existing earable authentication schemes generally exploit the unique sensing capabilities discussed above to verify user identity. These approaches can be broadly categorized into three paradigms: activity-based, interaction-based, and biometric-based methods.

1) Activity-Based Methods: Activity-based methods utilize dynamic features generated during users' daily activities. For example, EarDynamic [17] combines the static geometry of the ear canal with the dynamic motion characteristics of speech. EarPrint [6] analyzes sound conduction patterns from the throat to the ear canal, while MandiPass [5] captures unique jaw structure characteristics through vibrations induced by speech. EarPPG [7] extracts unique photoplethysmography (PPG) signals from blood vessel geometry changes during speech, and EarGate [9] extracts gait characteristics from bone-conducted sounds generated by foot impacts during walking. While effective in capturing dynamic features, these methods are sensitive to ambient noise and motion artifacts. For instance, speech-induced features can be masked by background noise, and gait signals may become distorted due to external vibrations.

2) Interaction-Based Methods: Interaction-based methods require users to perform predefined specific actions to complete authentication. For instance, TeethPass [10] and ToothSonic [43] leverage tooth movements, with the former relying on bone-conducted sounds generated by teeth occlusion, while the latter captures acoustic signals triggered by specific tooth gestures. Similarly, EarSlide [11] records acoustic fingerprints created by sliding a finger across the face, and BudsAuth [12] captures user-specific tissue features during finger-sliding movements. Despite achieving high recognition accuracy, these methods often face practical limitations such as poor user experience, high action complexity, and low authentication efficiency. In contrast, BaroAuth authenticates users through their natural speaking behavior, eliminating the need for additional cooperative actions, which improves the user experience and authentication efficiency.

3) Biometrics-Based Methods: Biometric feature-based methods rely on users' inherent biosignals, such as breathing, heartbeat, or ear canal acoustic properties, for authentication. BreathSign [13] captures in-ear sounds generated during breathing, HeartPrint [14] passively records heartbeat sounds, EarPass [15] extracts heartbeat patterns from blood pulse signals, and LR-Auth [16] analyzes the modulation characteristics of the ear canal to sound. While these methods demonstrate high accuracy in specific scenarios, they generally depend on specialized sensors, increasing hardware costs and design complexity. Additionally, the slow fluctuations of physiological signals, such as heartbeat and breathing, limit their suitability for real-time and high-frequency authentication.

VIII. Comparison With Existing Earable-Based Authentication Methods

We build a systematic comparison in Table VII, which catalogs representative earable-based authentication schemes across various dimensions.

Security: BaroAuth leverages barometric pressure variations inside the sealed ear canal during articulation. These SPSs arise from physiological canal deformation and cannot be reproduced by external audio playback or re-recorded speech, making the system naturally resistant to replay attacks. Our evaluation shows that knowledge-based imitation attacks increase the FAR when short commands (i.e., 2–5 word phrases) are used, FAR may rise to 2–2.5%, exceeding the stringent threshold (typically below 1%) required in high-security applications such as mobile payments [51], [52]. To approach this level, longer commands can be enforced, as 6–7 word phrases reduce FAR to nearly 1%, and multi-session authentication, where two distinct commands in consecutive sessions are required, further lowers the effective FAR to about 8×10^{-4}. Furthermore, when tracing back the experimental attack cases, we found that adversaries who succeeded with short commands were rarely observed to succeed with longer ones, which makes such multi-session policies more effective in practice. BaroAuth can thus be configured with longer or multi-session strategies to meet stricter security requirements, at the cost of additional user interaction. Also, prior studies (e.g., OnePiece [50]) have explored PVC or 3D-printed ear models, but replicas fail to capture the soft-tissue compliance and dynamic patterns of the human ear canal. Finally, although AI-driven synthesis may generate fake signals, SPSs encode anatomy-specific and fine-grained temporal dynamics that remain extremely difficult to mimic. Future deployments can further enhance robustness by integrating adversarial discriminators or perturbation detectors.

Usability: Compared with alternative modalities, BaroAuth shows distinct usability advantages. Voice-based methods degrade under noise, while BaroAuth remains stable as its signals stem from ear-canal deformation rather than external acoustics. IMU-based schemes (e.g., MandiPass [5], BudsAuth [12]) are sensitive to motion artifacts, and Jawthenticate [44] further

TABLE VII
COMPARISON WITH EXISTING EARABLE-BASED AUTHENTICATION APPROACHES

Approach	Sensor	Feature	Mode	Subject	C-f	N-p	F-p	A-s	C-d	SR	Power	Enroll	Latency	Performance
EarDynamic [17]	IEMs	Ear Canal Geometry	Active	24	✓	×	×	✓	×	44.1 kHz	-	-	-	93.04% Acc.
MandiPass [5]	IMU	Mandible Structure	Active	34	✓	✓	✓	✓	×	160 Hz	-	1 s	2000 ms	1.28% EER
EarGate [9]	IEMs	Gait Pattern	Passive	31	✓	✓	✓	✓	×	48k Hz	167.27 mJ	6 s	74.25 ms	3.23% FAR, 2.25% FRR
EarPPG [7]	PPG	Ear Canal Geometry	Passive	25	×	✓	✓	×	×	128 Hz	-	-	-	94.84% Acc.
Jawthenticate [44]	IMU	Jaw Structure	Passive	41	×	✓	✓	✓	✓	60 Hz	268 mJ	-	733 ms	97.07% BAC
EarPrint [6]	IEMs	Physiological Behaviors	Passive	50	✓	✓	×	✓	✓	2 kHz	1332 mJ	30 s	2255 ms	0.58% FAR, 10.9% FRR
Voice In Ear [45]	IEMs	Voiceprint	Passive	23	✓	✓	×	✓	✓	48 kHz	-	75 s	389 ms	3.64% EER
TeethPass [10]	IEMs	Teeth Structure	Active	22	✓	✓	×	✓	×	-	-	3.8 s	277 ms	96.8% Acc.
ToothSonic [43]	IEMs	Teeth Structure	Active	25	✓	✓	×	✓	×	-	-	-	-	1.1% FAR, 1.15% FRR
EarSlide [11]	IEMs	Fingerprint	Active	26	✓	✓	✓	✓	×	-	1100 mJ	81 s	359 ms	0.13% FAR, 3.14% FRR
BudsAuth [12]	IMU	Tissue Composition	Active	20	✓	✓	✓	✓	✓	200 Hz	3.16 mJ	5–6 s	1000 ms	5% EER
Nakamura et al. [46]	EEG	EEG Rhythms	Passive	15	×	✓	×	✓	×	1200 Hz	-	190 s	60 s	2.3% FAR
EarEcho [47]	IEMs	Ear Canal Geometry	Passive	20	✓	×	×	✓	×	44.1 kHz	114.7 mJ	1–3 s	181.4 ms	94.52% BAC
BreathSign [13]	IEMs	Breathing Pattern	Passive	20	✓	✓	×	✓	×	-	-	3–5 s	-	95.22% BAC
HeartPrint [14]	IEMs	Heartbeat Pattern	Passive	45	×	×	✓	✓	✓	-	1052.2 mJ	2–5 min	1576.3 ms	1.6% FAR, 1.8% FRR
EarPass [15]	PPG	Heartbeat Pattern	Passive	10	×	✓	✓	✓	×	100 Hz	-	-	-	98.7% Acc.
Hu et al. [48]	IEMs+OEMs	Ear Canal Geometry	Passive	22	✓	×	×	×	×	-	100.68 mJ	12 min	595.97 ms	4.84% EER
EarID [49]	IEMs	Physiological Behaviors	Passive	50	✓	×	×	✓	×	2 kHz	0.35mAh	-	2.1s	3.38% FAR, 4.89% FRR
LR-Auth [16]	IEMs+OEMs	Ear Canal Geometry	Active	30	✓	✓	✓	✓	×	44.1 kHz	2.5 mJ	1 s	100 ms	99.8% BAC
OnePiece [50]	OEMs	Ear Canal Geometry	Active	36	✓	✓	✓	✓	×	44.1 kHz	25.3mAh/h	5 min	138 ms	97% Acc.
BaroAuth (ours)	Barometer	Ear Canal Geometry	Passive	25	✓	✓	✓	✓	✓	18 Hz	131mJ	1min	220.42 ms	0.41% FAR, 1.23% FRR

Note: C-f = Contact-free (no extra contact beyond normal wearing); N-p = Noise-proof (robust against ambient acoustic or motion noise); F-p = Function-proof (does not interfere with normal audio/communication functions); A-s = Anti-spoofing (resistant to replay or several spoofing attacks); C-d = COTS-deployment (whether deployed on commercial mobile devices); SR = Sampling Rate; Acc. = Accuracy; BAC = Balanced Accuracy; FAR = False Acceptance Rate; FRR = False Rejection Rate.

requires a TMJ-mounted IMU plus a reference sensor, adding complexity and reducing comfort. Other IEM-based schemes (e.g., TeethPass [10], ToothSonic [43]) demand non-standard interactions, increasing learning cost. Ultrasound-based schemes (e.g., EarPrint [6], EarEcho [47]) not only consume high energy but also interfere with audio playback, compromising daily use. In contrast, BaroAuth simply embeds a small barometer into the earbud, enabling passive, low-cost, and unobtrusive sensing that preserves normal earbud functionality and ensures long-term comfort. A potential concern is whether audio playback interferes with authentication. As detailed in Supplementary Appendix D, SPSs recorded with 70% volume music playback remain consistent with those in silence, confirming that internal audio does not affect canal-pressure sensing or the reliability of BaroAuth. Enrollment is also lightweight: EarPrint [6] requires a 30 s continuous recording, and EarDynamic [] needs 15 fixed passphrases Blpar∼30–45 sBrpar. BaroAuth achieves comparable duration Blpar∼1 minBrpar but with fewer Blpar3–5Brpar and more natural commands,aligning with real-world usage. While MandiPass [5] enables near zero-cost enrollment with a single "EMM" utterance (0.2 s), this highly constrained input limits long-term practicality, whereas BaroAuth balances ease of enrollment with realistic usability.

Computation and energy overhead: On smartphones, BaroAuth achieves 220–400 ms end-to-end latency across devices (Table V). In a 2-hour continuous run (Fig. 22), the additional battery drain over a bluetooth-only baseline (6%) is 2.5%–6.0%, which corresponds to 69–218 mJ per authentication event. On earbuds, BaroAuth remains stable at low sampling rates (18 Hz), with per-bud power of 13.69 mW (and 17.39/20.72 mW at 50/108 Hz) in Table VI; combined with our commercial TWS battery survey, this enables ∼15 h of continuous operation at 18 Hz. In contrast, ultrasound/acoustic schemes typically require kHz-level sampling and report substantially higher event energy (e.g., EarPrint [6] ≈1.3 J), which challenges real-time mobile deployment.

Observable features: BaroAuth detects SPSs reflecting ear-canal deformation during speech. IMU-based approaches (e.g., MandiPass [5], BudsAuth [12], Jawthenticate [44]) only measure accelerometric data or bone-conduction vibrations, which cannot reflect the air-volume dynamics within the sealed ear canal. Ultrasound-based works (e.g., EarDynamic [17], EarEcho [47], Hu et al. [48], LR-Auth [16], OnePiece [50]) achieve similar sensing via echo changes. The key difference lies in coupling and cost: ultrasound reuses audio hardware with minimal overhead but is tied to the audio channel and requires high-rate processing, raising power and scheduling costs. BaroAuth instead uses a dedicated low-power barometer to directly capture low-frequency dynamics. Although this adds a small hardware cost, the sensor is compact (standard I^2C/SPI) and independent of audio. This decoupling, combined with low sampling rates (18–108 Hz), ensures BaroAuth's energy efficiency and deployability.

IX. DISCUSSION AND LIMITATIONS

Prototype integration and on-device processing: Our current Barobuds prototype primarily validates the sensing principle; the UI workflow and on-device processing are not yet production-grade or fully integrated with commercial earbuds. While this does not affect the sensing effectiveness, we plan to collaborate with manufacturers to integrate BaroAuth into mature platforms for seamless everyday use in strong-authentication scenarios (e.g., access control, payment-grade settings subject to platform policies).

Sensitivity to vigorous head movements: BaroAuth relies on a stable ear-canal seal. Under vigorous head movements, the earbud may loosen, degrading occlusion and signal integrity. Our EMD-based drift elimination is stable under natural motions; however, strong shaking can still introduce performance loss, which is a limitation shared by many in-ear systems that depend on sealing. Future work will add seal-quality detection, opportunistic authentication (triggering only when the seal is good), and hardware enhancements (e.g., optional ear hooks or foam tips).

Security–usability trade-off under knowledge-based imitation: As analyzed in Section VIII-A, when exposed to knowledge-based imitation, short commands (2–5 words) can push FAR above the $<1\%$ bar expected by payment-grade services (e.g., 2–2.5% in our tests). This behavior stems from the sensing modality and threat model and should be regarded as an inherent limitation rather than a tunable parameter. To reach the high-security regime, BaroAuth needs hardened modes (longer passphrases, e.g., 6–7 words, or two-session verification), which we explicitly acknowledge introduce non-negligible usability overhead.

Cohort scope, scale, and longitudinal robustness: Our data-collection cohort is male-heavy (84%), student-dominant (80%), and primarily native Chinese (76%), with few older and female participants (mean age 27.0 ± 7.21 yr; range 21–50). Together with the modest scale (25 participants, limited sessions), this constrains statistical power and external validity across demographics, languages, usage contexts, and devices. SPS characteristics may vary with ear-canal morphology, age-related biomechanics, prosodic rhythm, and earbud fit/occlusion stability; hence the current results are most representative of young student users. We will (i) enlarge and rebalance the cohort, including a broader age distribution, a more balanced gender ratio, recruitment beyond student groups, and greater representation of native English speakers; (ii) extend the study longitudinally and across devices; and (iii) provide a broader range of eartip sizes to better match diverse ear canals, together with quick leak checks during data collection. *Full cohort statistics are provided in Appendix G of the Supplementary Material.*

Interference from non-speech jaw movements: Non-speech jaw activities such as eating and drinking also induce pressure variations in the ear canals and can in principle interfere with BaroAuth. In our system, these activities do not autonomously initiate authentication, and they can affect the system only if they occur within the short window when the user is prompted to speak the passphrase. Appendix I presents a qualitative and quantitative analysis of this jaw-motion interference. The waveforms of drinking and eating exhibit isolated or highly periodic biphasic deformation cycles, which are clearly different from the continuous, syllable-like SPS segments induced by speaking the passphrase. In this supplementary study, we recruit eight participants and collect additional ear-canal pressure data while they eat and drink without speaking. When we treat these segments as jaw-motion impostor trials and feed them through the unchanged BaroAuth pipeline (pressure-only segmentation followed by BaroIDNet at the same operating point as in the main experiments), the 160 jaw-motion trials yield no false acceptances. This dataset is still limited in size and behavioral diversity, so extremely rare false acceptances caused by atypical jaw motions cannot be entirely ruled out. Future work will enlarge the jaw-motion corpus and train a lightweight front-end classifier, possibly augmented with inertial sensing, to reject non-SPS segments before they are passed to BaroIDNet.

X. Conclusion

In this paper, we propose BaroAuth, a novel biometric authentication system that utilizes SPSs captured by MEMS barometers in earbuds to extract dynamic and static physiological features, enabling efficient, robust, and accurate user authentication. By leveraging unique in-ear pressure signals, BaroAuth adapts to diverse mobile scenarios, withstands environmental noise, and resists both zero-knowledge and knowledge-based attacks. Extensive experiments demonstrate its effectiveness and robustness. However, BaroAuth has limitations, such as high FRR when the user shakes their head violently during speech. Future work will address this by using head motion data to compensate for ear pressure signal changes. Additionally, we plan to evaluate BaroAuth on various mobile devices and expand the user base to enhance generalization and applicability.

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